

Urban Risk Assessment: Urbanity Indicator, Risk Score, and Risk Category

Overview:

This documentation consolidates the theoretical foundation, implementation methodology, data requirements, and practical workflow for engineering three critical features in earthquake impact assessment: the **urbanity indicator**, **risk score**, and **risk category**. These features enable differentiated risk evaluation for urban versus rural earthquake events and support decision-making for disaster preparedness, urban planning, and emergency response.

1. Scientific Theory and Conceptual Framework

Fundamental Principles

Urban Earthquake Impact Differs from Rural Impact

Urban areas concentrate population, critical infrastructure, and economic assets. An earthquake of magnitude 5.0 in a densely populated city can cause significantly more damage, casualties, and economic loss than the same earthquake in a remote rural region. This principle necessitates location-aware risk assessment beyond magnitude and depth alone.

Urbanity as a Risk Multiplier

The urbanity of an earthquake location acts as a multiplicative factor on seismic hazard. Key drivers include:

- **Population density:** Higher population exposure amplifies casualty and impact potential.
- **Building stock vulnerability:** Urban areas often have mixed construction standards; older buildings are particularly vulnerable.
- **Infrastructure concentration:** Damage to power grids, water systems, transportation networks cascades rapidly in urban settings.

- **Economic exposure:** Urban property values are orders of magnitude higher, making economic loss more severe.

Physical Basis of Risk Scores

The urban risk score integrates:

- **Magnitude (M_w):** Total seismic energy released; a logarithmic measure where each unit increase represents ~ 31.6 times more energy.
- **Depth:** Shallow earthquakes dissipate less energy before reaching the surface, producing stronger ground shaking.
- **Urbanity indicator:** Binary classification (urban=1, rural=0) that modulates the risk calculation.

For urban risk, the relationship extends to:

Urban Risk Score Function

$$\text{Risk Score} = g(M, d, \text{urbanity indicator})$$

This acknowledges that urbanity modulates the final risk through exposure and vulnerability factors.

Practical Urban Risk Formula

A scientifically justified, practical urban risk score is:

For Urban Areas (urbanity_indicator = 1):

$$\text{Risk Score} = \min \left((M_w^2) \times \frac{1}{\max(d, 1)} \times 1.5, 10 \right)$$

For Rural Areas (urbanity_indicator = 0):

$$\text{Risk Score} = \min \left(M_w \times \frac{1}{\max(d, 1)}, 10 \right)$$

Justification:

- **Magnitude squaring for urban:** Captures non-linear increase in urban damage with magnitude. Larger earthquakes cause disproportionately more urban damage.
- **Reciprocal depth term:** Reflects physical attenuation; shallow earthquakes (low d) produce high ground motion and risk.
- **Urbanity multiplier (1.5):** Empirically calibrated factor reflecting 50% additional risk in urban contexts for equivalent seismic parameters.
- **Capping at 10:** Normalizes scores to a meaningful [0, 10] range for interpretability and prevents extreme outliers.

2. Data Source Considerations

Urbanity Indicator Data Requirements

Primary Source: Urban Boundary Polygons

The urbanity indicator requires geospatial data defining urban boundaries. Recommended sources:

Source	Coverage	Pros	Cons
OpenStreetMap (OSM)	Global	Free, regularly updated, community-maintained	Variability in urban definition across regions
Natural Earth	Global, coarse	Clean, curated datasets	Low resolution; may miss small cities
Government GIS Portals	National/regional	Authoritative, legally defined urban zones	Access varies; sometimes behind paywalls

Global Human Settlement Layer	Global	High-resolution, satellite-derived	Academic/research focus; licensing may apply
World Bank Urban Data	Global	Aligned with development indicators	Lower spatial resolution

Data Format: Shapefiles (.shp, .shx, .dbf) or GeoJSON files containing polygon geometries of urban areas.

Coordinate Reference System (CRS): Ensure urban boundary data uses WGS84 (EPSG:4326) or is reprojected to match earthquake coordinates.

Magnitude and Depth Data Requirements

For accurate urban risk score calculation, use:

- **Magnitude:** Prioritize Moment Magnitude (Mw) or variants (mwc, mww, mwb, mwr).
- **Depth:** Authoritative sources (USGS, ISC) provide refined depths after initial reporting.

Quality Considerations:

- Avoid saturated magnitude scales (mb, Ms) for large earthquakes ($M > 6.5$).
- Document magnitude sources and conversions for reproducibility.
- Use consistent depth databases to prevent artificial variability.

3. Feature Definitions and Specifications

Feature 1: Urbanity Indicator

Aspect	Specification
Data Type	Binary (0 or 1)

Definition	1 if earthquake epicenter lies within an urban boundary polygon; 0 otherwise
Derivation Method	Spatial join (point-in-polygon) using GIS operations
Geographic Reference	WGS84 (EPSG:4326) latitude/longitude
Null Handling	Assign 0 if spatial join returns null (point outside any urban polygon)
Temporal Variability	Static per earthquake; urban boundaries evolve over decades (use snapshot consistent with earthquake era if possible)

Feature 2: Urban Risk Score

Aspect	Specification
Data Type	Continuous, float [0, 10]
Definition	Quantitative measure of earthquake impact risk, differentiated by urbanity
Formula (Urban)	$(M_w^2) \times (1 / \max(d, 1)) \times 1.5$, capped at 10
Formula (Rural)	$M_w \times (1 / \max(d, 1))$, capped at 10
Input Variables	Mw (Moment Magnitude), d (depth in km), urbanity_indicator
Normalization	Min=0 (no earthquake), Max=10 (extreme urban earthquake)
Interpretation	Higher scores indicate greater earthquake impact potential

Feature 3: Risk Category

Aspect	Specification
Data Type	Categorical (ordinal)
Categories	Low, Moderate, High, Very High
Thresholds	Very High: score ≥ 7.5 ; High: $5.0 \leq \text{score} < 7.5$; Moderate: $2.5 \leq \text{score} < 5.0$; Low: score < 2.5
Derivation	Discretization of urban_risk_score using predefined thresholds
Encoding for ML	Ordinal: Low=1, Moderate=2, High=3, Very High=4 (reflects hierarchical order)
Use Case	Actionable categorization for alerts, policy, and resource allocation

4. Challenges and Solutions

Challenge: Handling Imbalance During Model Training

Problem: Regression models trained on imbalanced data may overfit to minority classes (urban samples) or ignore them entirely, leading to biased predictions and poor generalization.

Solutions:

- Regularization:** Use models with built-in penalties such as Ridge, Lasso, Elastic Net, or tree-based methods with controls like max_depth and subsample to reduce overfitting risk.

- 2. **Cross-validation:** Employ 5-fold stratified cross-validation to ensure balanced representation of both urban and rural subsets in training and validation splits, enhancing robustness.
- 3. **Sample Weights:** Compute sample weights inversely proportional to the frequency of urbanity and risk category classes. Assign these weights during model fitting to emphasize underrepresented urban samples without synthetic oversampling.
- 4. **Evaluation Discipline:** Monitor the difference between training and test R^2 scores; if the train score exceeds the test by more than 0.1, it indicates overfitting. Adjust model complexity accordingly.

5. Recommended Workflow for Feature Engineering

Workflow Summary

Step	Task	Output
1	Acquire urban boundary shapefile (EPSG:4326)	urban_areas_full.shp
2	Convert earthquake data to GeoDataFrame	gdf_points with geometry
3	Perform spatial join (point-in-polygon)	gdf_joined with index_right column
4	Create urbanity_indicator from spatial join result	Binary feature (0/1)
5	Calculate urban_risk_score using adaptive formula	Continuous feature (0–10)

6	Assign risk_category via threshold binning	Categorical feature (Low/Moderate/High/Very High)
7	Encode risk_category ordinally for ML	Numeric feature (1–4)
8	Quality control: check ranges, distributions, correlations	Validated dataset
9	Merge features back to original DataFrame	Final preprocessed dataset
10	Document data sources, CRS, magnitude versions, formula rationale	Reproducible pipeline

6. Summary

The **urbanity indicator**, **risk score**, and **risk category** are scientifically grounded, domain-informed features that enable differentiated earthquake impact assessment. By leveraging geospatial data and adaptive risk formulas, they capture the amplified vulnerability of urban areas while maintaining interpretability and avoiding common overfitting pitfalls. Implementation requires careful data sourcing, spatial analysis, and regularized modeling techniques, but yields a robust framework for urban earthquake risk prediction and decision support.